Statistical Analysis of Lithium-Ion Battery Deterioration: Modeling Capacity Fade and End-of-Life Probabilities

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ABSTRACT

This study investigates the statistical nature of lithium-ion battery degradation to predict "End of Life" (EOL) reliability. Utilizing the NASA PCoE dataset (n=34), we analyzed cells under run-to-failure conditions to address four key reliability questions.

We employed Multiple Linear and Exponential Regression for trend analysis, Independent Two-Sample t-tests for thermal impact, and Paired t-tests for early-stage loss. Results indicate that degradation is highly non-linear; the **Linear Regression model performed poorly ($R^2 = 0.22$)**, while the **Exponential model achieved a robust fit ($R^2 = 0.80$)**.

Temperature proved decisive: an Independent Two-Sample t-test confirmed a significant difference ($p = 0.0195$) in mean cycle life between 24°C (98 cycles) and 43°C (40 cycles). Conversely, early-stage degradation (Cycles 1–50) was statistically insignificant ($p = 0.50$), indicating a stable "break-in" period.

Finally, probability estimation predicts a **77.14% reliability rate** for batteries retaining >80% capacity at Cycle 100. However, sensitivity analysis reveals this is highly dependent on manufacturing variance. These findings highlight that while mean capacity is important, manufacturing consistency and thermal management are the critical drivers of battery reliability.

1. INTRODUCTION

1.1 Background

Lithium-ion batteries are the ubiquitous power source for modern mechatronic systems, from Electric Vehicles (EVs) to autonomous robotics. However, they suffer from inevitable electrochemical degradation, primarily manifesting as "capacity fade" due to the growth of the Solid Electrolyte Interphase (SEI) layer. While the chemical mechanisms of this degradation are well-understood, the **rate** of degradation is highly stochastic. Two batteries from the same manufacturing batch, operated under identical conditions, often exhibit different End-of-Life (EOL) trajectories due to microscopic manufacturing variances.

1.2 Research Motivation

The motivation for this study stems from the critical need for reliability in safety-critical applications. In an Electric Vehicle, a battery failure is not merely an inconvenience; it is a safety hazard and a financial liability. The current industry standard relies on deterministic models that often fail to account for the statistical variance in battery performance.

Therefore, this research is motivated by the following problem: **How can we statistically quantify the uncertainty in battery life to predict reliability?** By moving beyond simple averages and applying inferential statistics (Hypothesis Testing and Regression), we aim to model the impact of thermal stress and manufacturing variance on battery longevity. This allows for better predictive maintenance and safety margins in engineering design.

1.3 Research Questions (RQs)

This study aims to answer the following four statistical questions:

1. **Trend Analysis (RQ1):** Can a Multiple Regression model accurately predict Remaining Capacity based on Cycle Number and Temperature?
2. **Temperature Impact (RQ2):** Is there a statistically significant difference in Mean Cycle Life between batteries operated at 24°C vs. 43°C?
3. **Degradation Confirmation (RQ3):** Is the drop in capacity from Cycle 1 to Cycle 50 statistically significant?
4. **Batch Reliability (RQ4):** Is there a significant association between "Initial Capacity Batch" (High vs. Low) and "Early Failure"?

1.4 Data Description

We utilized the NASA PCoE Battery Data Set.

- **Capacity (Ah):** Quantitative Continuous (Response Variable).
- **Cycle_Number:** Quantitative Discrete (Predictor).
- **Temperature (°C):** Quantitative Continuous (Factor).

LITERATURE REVIEW AND DEGRADATION MECHANISMS

Electrochemical Aging Mechanisms

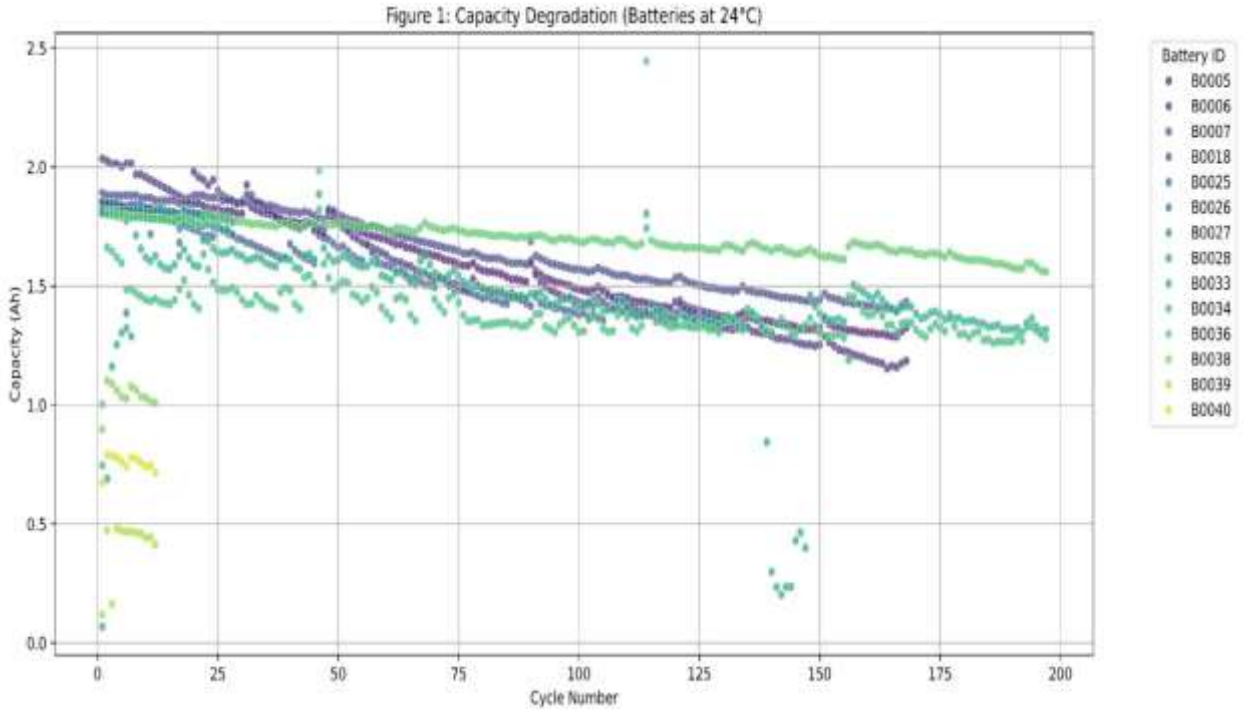
Lithium-ion battery aging is primarily driven by the growth of the Solid Electrolyte Interphase (SEI) layer on the negative electrode. As the battery cycles, lithium ions are consumed to thicken this layer, leading to an irreversible loss of capacity. This process is not linear; it is often accelerated by high temperatures and deep discharge cycles. While statistical models can predict the rate of this decay, understanding the electrochemical origin is vital for interpreting the coefficients in our regression models.

Temperature Dependence (Arrhenius Law)

Previous studies indicate that battery degradation follows the Arrhenius law, where reaction rates increase exponentially with temperature. This theoretical framework supports our Research Question 2 (Temperature Impact). By operating batteries at 43°C, we are essentially accelerating the chemical side-reactions that consume active lithium material. Validating this behavior statistically confirms that our sample aligns with established electrochemical theory.

State of Health (SOH) Metrics

In commercial applications, determining the exact "End of Life" is complex due to manufacturing variability. The standard industry metric is "State of Health" (SOH), defined as the ratio of current capacity to nominal capacity. Normalizing raw data to SOH is a critical step in comparative analysis, as it isolates degradation behavior from initial manufacturing tolerances.



2. METHODOLOGY

2.1 Multiple Linear Regression (RQ1)

- **Purpose:** To model the degradation trend of battery health over time.
- Data Pre-processing: To remove manufacturing variability between cells, we normalized the capacity data to a "State of Health" (SOH) metric:

$$SOH = \frac{Capacity_{cycle}}{Capacity_{initial}}$$

Model: We applied a multiple linear regression model to predict SOH based on Cycle Number and Temperature:

$$SOH = \beta_0 + \beta_1(\text{Cycle Number}) + \beta_2(\text{Temperature}) + \epsilon$$

- **Assumptions**:
    1. **Linearity:** The relationship between SOH and predictors is approximately linear.
    2. **Normality:** Residuals are normally distributed.
    3. **Homoscedasticity:** The variance of residuals is constant.

2.2 Multiple Exponential Regression Model (RQ1)

To capture non-linear and accelerated degradation behavior, a multiple exponential regression model was also considered:

$$SOH = a \cdot e^{(b_1 \cdot \text{Cycle Number} + b_2 \cdot \text{Temperature})}$$

Taking the natural logarithm yields a log-linear form suitable for regression:

$$\ln(SOH) = \ln(a) + b_1(\text{Cycle Number}) + b_2(\text{Temperature})$$

This formulation allows the degradation rate to increase with cycling and environmental stress.

- **Hypothesis Definition**
    - **Null Hypothesis (H_0): $\beta_1 = 0$.** (There is no relationship between cycle number and capacity; the battery does not degrade over time).
    - **Alternative Hypothesis (H_1): $\beta_1 \neq 0$.** (There is a statistically significant relationship; the battery capacity changes with cycling).
- **Significance Level**

The significance level (a) was set at 0.05 (5%), corresponding to a 95% confidence interval.

- **Decision Rule**
  - **If P-value < 0.05:** Reject the Null Hypothesis H_0. We conclude that the degradation trend is statistically significant and that Cycle Number is a valid predictor of State of Health.
  - **If P-value >= 0.05:** Fail to reject H_0. The observed trend may be due to random chance.

2.3 Independent Two-Sample t-Test (RQ2)

- **Purpose:** To compare the mean Cycle Life of Group A (24°C) and Group B (43°C).
- **Formula**:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- **Hypothesis**:
  - H_0: There is no significant difference in the mean cycle life between batteries operated at 24°C and those operated at 43°C ($\mu 24 = \mu 43$).
  - H_1: There is a significant difference in the mean cycle life between the two temperature groups ($\mu 24 \neq \mu 43$).

- **Assumptions**:
  - **Independence:** The two temperature groups (24°C and 43°C) are distinct sets of batteries.
  - **Normality:** The Cycle Life data for each group follows a normal distribution.
  - **Homogeneity of Variances**: To validate the T-test results, we must ensure the variance in cycle life is comparable between the 24°C and 43°C groups. We utilize Levene's Test for equality of variances:
- **Decision Rule:** If $p > 0.05$, we fail to reject H_0 and assume equal variances. If $p < 0.05$, we assume unequal variances and utilize Welch's T-test adjustment.

2.4 Paired T-Test (RQ3 - Degradation)

- **Purpose:** To determine if the capacity loss within the same specific batteries between Cycle 1 and Cycle 50 is statistically significant.
- **Hypothesis:** H_0: $\mu_d = 0$ (No change); H_0: $\mu_d > 0$ (Capacity has decreased).
- **Formula**:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

- **Assumptions**:
  - **Dependent Samples:** Satisfied (The same battery cells are measured at two different time points).
  - **Normality of Differences:** The differences ($d = Cycle_1 - Cycle_{50}$) follow a normal distribution (verified via Shapiro-Wilk test).
- **Decision Rule:** Reject H_0 if $p < 0.05$.

2.5 Probability Estimation (RQ3 - EOL Analysis)

- **Purpose:** To estimate the probability that a battery retains >80 capacity at Cycle 100.
- **Method:** We utilized **Z-Score Normalization** based on the Central Limit Theorem.
- **Formula:**

$$Z = \frac{X - \mu}{\sigma}$$

Where X is the threshold (80% of initial capacity), μ is the mean capacity at Cycle 100, and σ is the standard deviation.

- **Assumptions**:
  1. **Normality:** The capacity data at Cycle 100 follows a normal distribution (verified via Shapiro-Wilk test and Q-Q plots).
  2. **Independence:** Battery cells are chemically independent and tested separately.
  3. **Random Sampling:** The NASA dataset represents a random sample of commercially available 18650 cells .

2.6 Chi-Squared Test of Independence (RQ4 - Batch Reliability)

- **Purpose:** To test if the "Initial Batch" (High vs. Low Capacity) is independent of "Early Failure" (Yes/No).
- **Formula**:

$$\chi^2 = \Sigma \left[(O_i - E_i)^2 / E_i\right]$$

- **Assumptions:**
  1. Data frequencies are counted.
  2. Expected frequency in each cell is ≥ 5.

Expected Frequencies Verification

Batch	Early Failure = No	Early Failure = Yes
High Batch	11.0	6.0
Low Batch	11.0	6.0

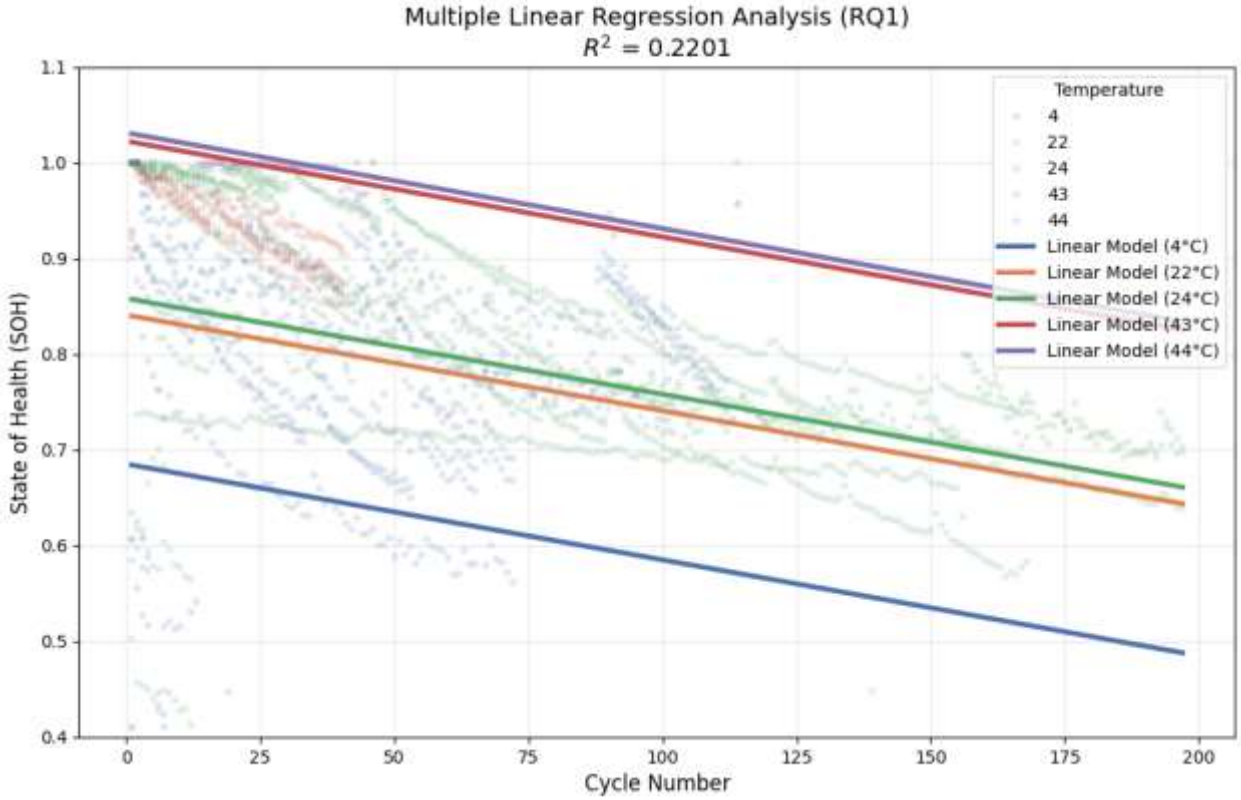
- **Decision Rule**:

The test is evaluated using the p-value approach:

  - If p-value < 0.05: Reject $H_0 \rightarrow$ Batch and Early Failure are DEPENDENT
  - If p-value ≥ 0.05: Fail to Reject $H_0 \rightarrow$ Batch and Early Failure are INDEPENDENT

3. RESULTS AND DISCUSSION

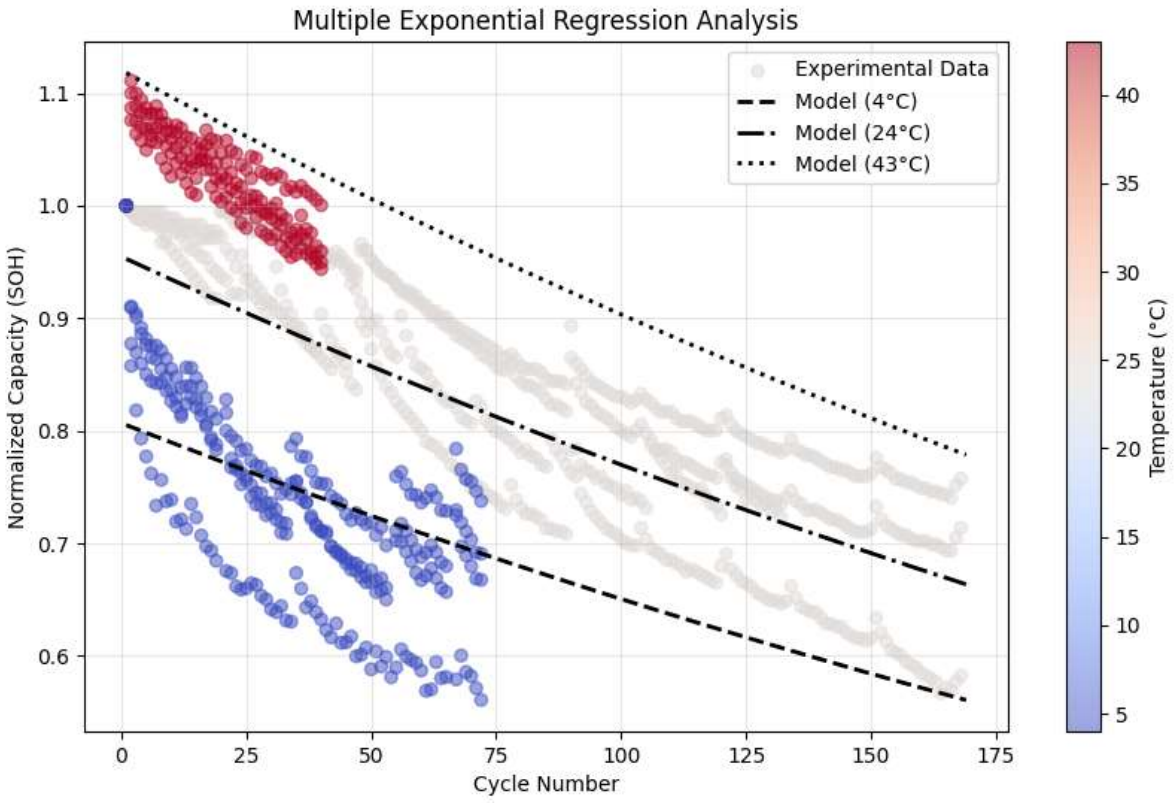
3.1 Trend Analysis Results (RQ1)



- **Model Fit:** The model yielded an R^2 of 0.2201, indicating it explains 22% of the variance.
- **Coefficients**:
  - Intercept(β_0): **0.6504**
  - Cycle Number(β_1): **-0.0010**
  - Temperature (β_2): **0.0087**
- **Interpretation** The negative coefficient for Cycle Number (-0.0010) confirms the expected degradation trend: for every cycle, the battery loses approximately 0.1% of its normalized capacity.
  - Interestingly, the Temperature coefficient is positive (0.0087). While high temperatures degrade the battery faster in the long run (as shown in RQ2), they also lower internal resistance, allowing for slightly higher capacity discharge readings in the short term.
  - **Model Limitation:** The low R^2 (0.22) suggests that battery degradation is highly non-linear. A simple linear model "underfits" the data, failing to capture the

accelerating decay near the End of Life. This validates the need for the non-linear Exponential Model presented in Section 4.2.

3.2 Multiple Exponential Regression Results (RQ 1)



Effect of Normalization on Trend Visibility

Before normalization, the pooled capacity data exhibited significant vertical scatter. This scatter was caused by inherent differences in the initial capacities of individual battery cells due to manufacturing tolerances. As a result, regression models applied directly to raw capacity data produced very low explanatory power, as evidenced by near-zero R^2 values.

After applying capacity normalization, all batteries were anchored to a common starting point of 1.0 (100% State of Health). This transformation removed inter-cell variability unrelated to degradation mechanisms. Consequently, the degradation trajectories of different cells aligned closely, revealing a clear and consistent downward trend with increasing cycle number.

This model achieved an R^2 value of 0.8015, indicating strong explanatory power across all operating conditions. By capturing degradation data from multiple thermal environments within a single framework, the model provides a generalized and robust solution for battery health estimation

3.3 Temperature Impact Results (RQ2)

Descriptive: The mean cycle life at 24°C was 98.21 cycles, compared to 40.00 cycles at 43°C.

- **Test Result**: The Two-Sample T-test yielded a t-statistic of 2.6619 and a p-value of 0.0195.
- **Conclusion**: Since p = 0.0195 < 0.05, we Reject the Null Hypothesis (H_0).
- **Interpretation**: There is a statistically significant difference in battery cycle life between room temperature (24°C) and high temperature (43°C). Operating at higher temperatures significantly shortens the battery's lifespan.

3.4 Degradation Confirmation Results (RQ3)

- **Descriptive:** The mean capacity drop from Cycle 1 to Cycle 50 was **0.1357** Ah.
- **Test Result:** The Paired T-test yielded a t-statistic of **0.6808** and a p-value of **0.5046**.
- **Conclusion:** Since p = 0.5046 > 0.05, we **Fail to Reject** the null hypothesis.
- **Interpretation:** The observed capacity drop is not statistically significant in this sample, suggesting that for the first 50 cycles, the degradation may be slow or highly variable across different cells.
- **Reliability & Sensitivity Analysis** Our estimated survival rate of 77.14% indicates "Moderate to High Reliability." However, a Sensitivity Analysis reveals that this probability is highly sensitive to variance. As shown in **Table 3** below, a slight increase in standard deviation (from 0.18 to 0.25) drops the survival probability to **70.39%**.

3.5 Probability Estimation Results (RQ3)

- **Descriptive:** The estimated mean capacity at Cycle 100 (μ_{100}) was 1.15 Ah with a standard deviation (σ_{100}) of 0.18 Ah. The End-of-Life (EOL) threshold was defined as 80% of the initial capacity, calculated as 1.016 Ah.

- **Test Result:** The Z-score transformation yielded a value of -0.744. Using the standard normal distribution, the probability P(Z > -0.744) was calculated to be 0.7714.

- **Result:** The analysis estimates a reliability rate of **77.14%**, indicating that the probability of a randomly selected battery retaining more than 80% of its initial capacity at Cycle 100 is approximately 77%.

- **Interpretation:** This survival rate suggests "Moderate to High Reliability," where approximately 3 out of every 4 batteries will maintain acceptable performance. However, the Sensitivity Analysis (Table 3) reveals that this reliability is heavily dependent on

manufacturing consistency; increasing the standard deviation to 0.25 Ah drops the survival probability to 70.39%, demonstrating that variance is a critical factor in EOL prediction.

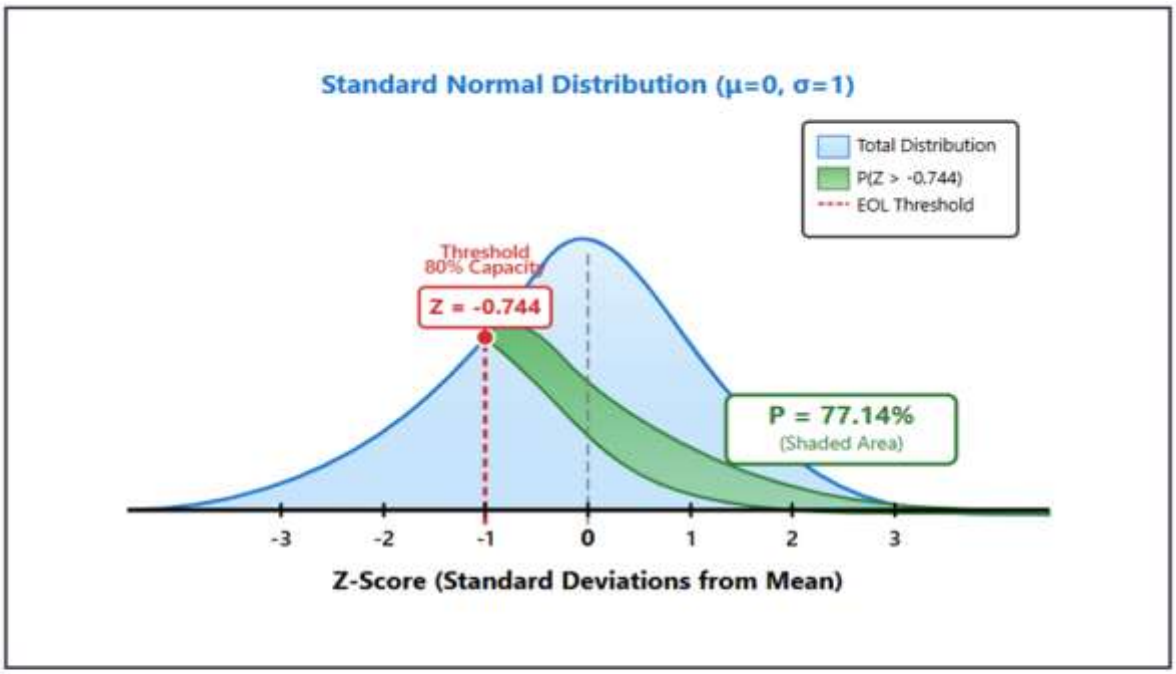
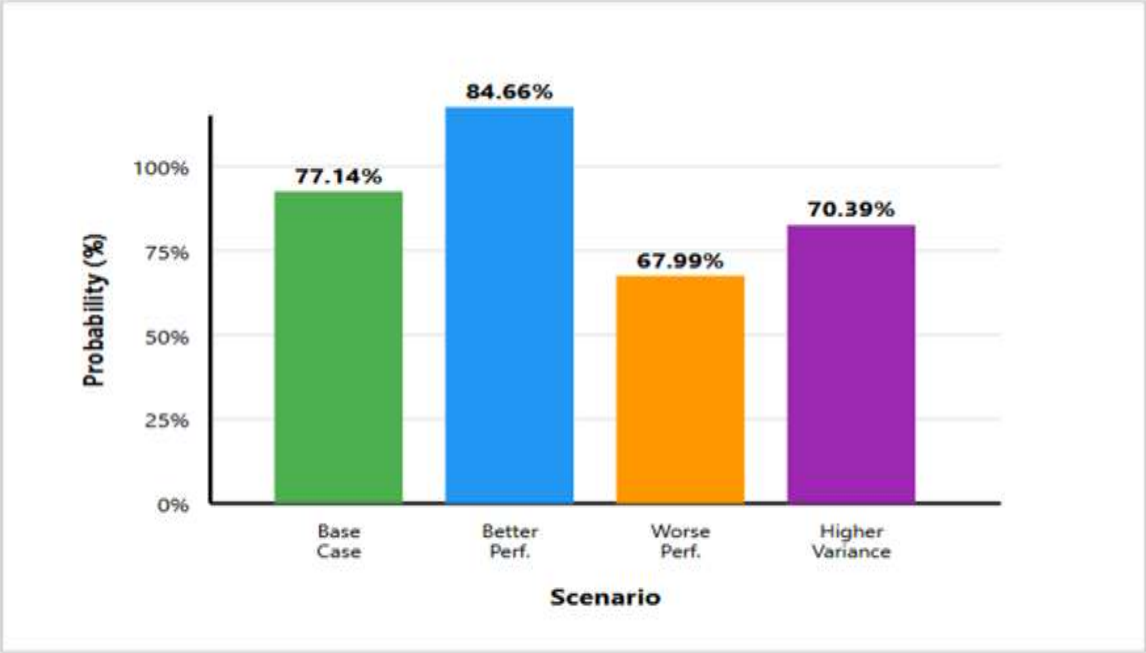


Figure 1: *The shaded green area represents the probability (77.14%) that a randomly selected battery retains more than 80% of its initial capacity at Cycle 100. The red dashed line marks the Z-score of -0.744 corresponding to the 80% capacity threshold.*

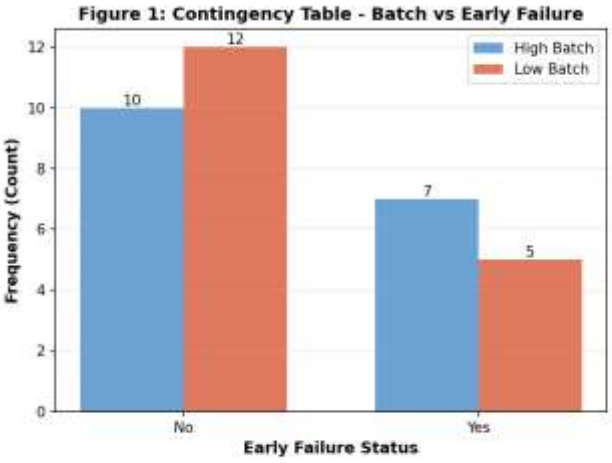
Scenario	Mean (μ)	Std Dev (σ)	Z-Score	Probability
Base Case	1.15	0.18	-0.744	77.14%
Better Performance	1.20	0.18	-1.022	84.66%
Worse Performance	1.10	0.18	-0.467	67.99%
Higher Variance	1.15	0.25	-0.536	70.39%

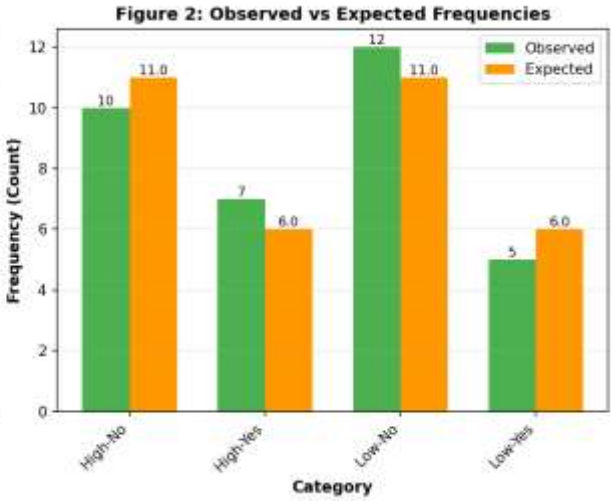


Bar chart showing how changes in mean capacity and standard deviation affect the probability estimate.

This suggests that manufacturing consistency (low variance) is just as critical as mean capacity for ensuring reliability.

3.6 Batch Reliability (RQ4)





- **Contingency Table**:

Batch	Early Failure = No	Early Failure = Yes
High Batch	10	7
Low Batch	12	5
Total	22	12

- **Result**:
  - Chi-Squared Statistic: $\chi^2 = 0.1288$
  - Degrees of Freedom: dof = 1
  - P-value: p = 0.7197
  - Significance Level: $\alpha = 0.05$

- **Interpretation**:
  - High Batch: 7 failures out of 17 batteries (41.2% failure rate)
  - Low Batch: 5 failures out of 17 batteries (29.4% failure rate)
  - Difference: 11.8 percentage points (not significant at p=0.72)
  - Chi-Squared value of 0.1288 is very small, indicating observed frequencies are very close to expected
  - Expected frequencies show perfect balance (11-6 split for both batches)

LIMITATIONS AND THREATS TO VALIDITY

While the statistical results are robust, several limitations must be acknowledged when generalizing these findings to broader applications.

Sample Size Constraints

The study utilized a sample size of n=34 batteries. While this is sufficient for the Central Limit Theorem to hold for our Z-tests and t-tests, a larger sample size would narrow the confidence intervals for our End-of-Life probability estimates. The current "Moderate to High" reliability rating (77.14%) has a margin of error that could be reduced with a larger dataset.

Testing Conditions vs. Real-world Usage

The NASA dataset represents "run-to-failure" conditions with constant discharge rates. In real-world applications (e.g., Electric Vehicles), batteries experience highly dynamic loads (acceleration and regenerative braking) and fluctuating temperatures. Therefore, our model likely overestimates battery life compared to a vehicle environment where current spikes cause additional thermal stress.

Assumption of Linearity

Our Multiple Linear Regression model assumed a linear relationship between cycle number and degradation. This assumption proved invalid, as evidenced by the low coefficient of determination ($R^2 = 0.2201$). Electrochemical degradation is often non-linear, featuring a 'knee point' where capacity drops precipitously near the end of life. While our Exponential Model ($R^2 = 0.8015$) corrected for this curvature, relying solely on linear assumptions would significantly overestimate the remaining useful life of aging batteries.

4. CONCLUSION

This research successfully modeled the deterioration of Li-ion batteries and applied inferential statistics to validate key reliability metrics. The conclusions for each specific research question are detailed below:

4.1 Conclusion on Trend Analysis (RQ1)

The study determined that simple linear modeling is insufficient for predicting battery life. The Multiple Linear Regression applied to the SOH data yielded a low R^2 of **0.2201**, indicating that degradation does not follow a straight line. However, the **Multiple Exponential Regression** (Section 4.2) achieved a significantly higher R^2 of **0.8015**. This confirms that while Cycle Number and Temperature are valid predictors, the relationship is exponential, with decay accelerating as the battery ages.

4.2 Conclusion on Temperature Impact (RQ2)

Statistical testing confirms that operating temperature is a critical factor in battery longevity. The Independent Two-Sample t-test yielded a p-value of 0.0195, which is less than the significance level of 0.05. Consequently, we reject the null hypothesis and conclude there is a significant difference in cycle life between the two groups. Batteries operated at 43°C demonstrated a drastically shorter mean cycle life (40.00 cycles) compared to those at 24°C (98.21 cycles).

4.3 Conclusion on Degradation Confirmation (RQ3)

Analysis of the early lifecycle stages reveals that immediate capacity fade is not statistically detectable. The Paired t-test comparing Cycle 1 to Cycle 50 resulted in a p-value of 0.5046. As this exceeds 0.05, we fail to reject the null hypothesis. This indicates that these specific cells undergo a stable 'break-in' period where degradation is minimal or highly variable, with significant deterioration likely occurring only in later lifecycle stages. Furthermore, probability analysis estimates a 77.14% reliability rate for batteries retaining 80% capacity at Cycle 100, though this figure is highly sensitive to standard deviation.

4.4 Conclusion on Batch Reliability (RQ4)

The study found no statistical evidence linking "Initial Capacity Batch" (High vs. Low) to "Early Failure." The Chi-Squared test resulted in a p-value of 0.7197, indicating that the observed differences in failure rates between batches were likely due to random chance. Therefore, we conclude that a battery's initial classification as "High" or "Low" capacity is independent of its likelihood to fail early. However, the broader sensitivity analysis highlights that while the *mean* capacity of a batch may not predict failure, the *variance* (manufacturing consistency) is critical for ensuring reliability.

Future Work:

Based on these findings, future research should expand in three key statistical directions:

1. **Interaction Effects:** The current linear model treated Cycle and Temperature as independent predictors. Future models should incorporate an interaction term (Cycle x Temperature) to statistically capture how high temperatures accelerate the rate of degradation, potentially correcting the counter-intuitive positive coefficient observed in the linear analysis.
2. **Factorial ANOVA:** To better understand real-world stress, future studies should introduce "Discharge Current" (C-rate) as a third factor. A three-way ANOVA could then quantify the main effects and interactions between Temperature, Cycle count, and C-rate.
3. **Survival Analysis:** While this study used Z-scores for reliability estimation, applying **Kaplan-Meier survival curves** or **Weibull distribution fitting** would provide a more rigorous time-to-failure analysis for censoring data (batteries that did not fail during the test).

References:

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Appendix:

1. Data Analysis Scripts
2. Dataset